Yongkang Li

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EDUCATION

Purdue University West Lafayette, IN
Ph.D. Student · Advisor: Prof. Yiheng Feng 2025 – Present

RESEARCH INTERESTS

Autonomous Driving, Connected and Automated Vehicles, Trajectory Prediction, Autonomous Vehicle Safety, World Models, Computer Vision

PROJECTS

Work Zone Risk Prediction Purdue University · with INDOT
Integrated Smart Work Zone project 2025 – Present

- Developing a roadside risk-warning system that protects workers in highway work zones. A two-stage pipeline first flags high-risk vehicles by time-to-collision (TTC), then a deep-learning trajectory-prediction model decides whether each flagged vehicle will intrude into the work area.
- Collected roadside radar data at multiple work-zone sites and used it to calibrate and refine the warning system.
- Deployed the system on an edge computer for real-time operation at active work zones during several field tests in Indiana.

Connected Vehicles Purdue University · with INDOT
Integrated Smart Work Zone project 2026 – Present

- Built a C-V2X warning system that brings work-zone alerts into the vehicle: a roadside unit (RSU) broadcasts work-zone information, and an onboard unit (OBU) receives it and shows it to the driver.
- Developed its two interfaces: an in-vehicle display that warns of the work zone ahead, workers on site and the reduced speed limit, and a digital-twin console that sets up the work zone and sends it to the RSU.
- Deployed the system on the lab's autonomous test vehicle and field-tested it at a real site with INDOT.

RESEARCH EXPERIENCE

Purdue University — CART Lab West Lafayette, IN
Graduate Research Assistant, Advisor: Prof. Yiheng Feng 2025 – Present

- **Trace2Map.** A conditional VAE that reconstructs a work zone's drivable-area heatmap in near real time from only about 10 vehicle trajectories, without road geometry or a manually annotated HD map. Used in place of the HD map in the multimodal predictor Trajectron++, the heatmap makes predictions safer (63–74% fewer drivable-area violations). Validated in SUMO, on human-driven trajectories in CARLA, and on a real work zone.
- **GenTFM.** A trajectory foundation model whose Variational Orthogonal Subspace Decomposition (VOSD) separates shared motion patterns from scene context and driving style, so one model learns from 25 heterogeneous datasets (33.5M scenes from 8 countries, e.g., Waymo, nuPlan, Argoverse 2, highD). It outperforms baselines such as MotionLM and generalizes to unseen datasets with about half their accuracy loss.
- **FINDAR.** Classifies adversarial attacks on trajectory prediction from features of the frozen GenTFM model, separating clean scenes from eight attack families (e.g., white-box PGD, black-box collision, gradual drift).

University of Macau — State Key Laboratory of Internet of Things for Smart City Macau, China
Research Assistant, Supervisors: Prof. Zhenning Li and Prof. Chengzhong Xu Jul. 2023 – Aug. 2024

- **Cognitively inspired trajectory prediction.** Developed a teacher–student model that mimics how human drivers observe and reason, with graph attention and Transformers for agent interactions, reducing prediction error by 15.5% with 30% fewer parameters (T-IV, ECAI).
- **Accident anticipation.** Added monocular depth cues to 3D modeling of dashcam video (+4.0% precision), and built a three-stage framework with LLM scene prompting that predicts when, where and what kind of accident will occur: +14.6% precision and 16.4% earlier anticipation (ACM MM oral, AAP).

Peking University — CIS Lab Beijing, China
Research Assistant, Supervisor: Prof. Yuesheng Zhu · multimodal deepfake and diffusion-image detection May – Jun. 2023

Sun Yat-sen University Shenzhen, China
Research Assistant, Supervisor: Prof. Wenyuan Yang · capsule-network deepfake detection Aug. 2022 – Apr. 2023

PUBLICATIONS

[1] H. Liao, H. Shen, B. Wang, **Y. Li**, Y. Tang, C. Wang, D. Zhuang, K. Chen, et al. **Think Before You Drive: World Model-Inspired Multimodal Grounding for Autonomous Vehicles**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[2] H. Liao*, **Y. Li***, Z. Li, C. Wang, Y. Guan, K. H. Tam, C. Tian, L. Li, C. Xu. **When, Where, and What? A Benchmark for Accident Anticipation and Localization with Large Language Models**. *ACM International Conference on Multimedia (ACM MM)*, 2024. **(Oral)**

[3] H. Liao*, **Y. Li***, Z. Li, C. Wang, C. Tian, Y. Huang, Z. Bian, K. Zhu, G. Li, Z. Pu, et al. **Less is More: Efficient Brain-Inspired Learning for Autonomous Driving Trajectory Prediction**. *European Conference on Artificial Intelligence (ECAI)*, 2024.

[4] H. Liao*, **Y. Li***, Z. Li, C. Wang, Z. Cui, S. E. Li, C. Xu. **A Cognitive-Based Trajectory Prediction Approach for Autonomous Driving**. *IEEE Transactions on Intelligent Vehicles (T-IV)*, 2024.

[5] H. Liao*, **Y. Li***, Z. Li, Z. Bian, J. Lee, Z. Cui, G. Zhang, C. Xu. **Real-Time Accident Anticipation for Autonomous Driving through Monocular Depth-Enhanced 3D Modeling**. *Accident Analysis & Prevention*, vol. 207, 107760, 2024.

[6] H. Liao, S. Liu, **Y. Li**, Z. Li, C. Wang, B. Wang, Y. Guan, C. Xu. **Human Observation-Inspired Trajectory Prediction for Autonomous Driving in Mixed-Autonomy Traffic Environments**. *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

HONORS AND AWARDS

Two Grand Prizes (ranked 1st and 2nd) and a 2nd Prize — China College Computer Competition (national) 2023, 2024
2nd Prize — China Students Service Outsourcing Innovation and Entrepreneurship Competition (national) 2023, 2024
2nd Prize, ranked 12th — Pangu Stone Cup Electronic Digital Forensics Competition (national) 2023
3rd Prize — China College Student Computer Design Competition (national) 2022
Registered software copyright — web link positioning and security detection system based on ChatGLM 2024

SKILLS

Programming Python, PyTorch, TypeScript / JavaScript, React, FastAPI, Linux, LaTeX
Systems Roadside radar and camera data, real-time edge deployment, C-V2X (SAE J2735 messages)
Design UI design, data visualization, video editing

LEADERSHIP AND SERVICE

LingRui Studio — Vice-President and UI Team Leader Oct. 2022 – Oct. 2023
Weiyang Volunteer Teaching Group — Head of Publicity and volunteer teacher in rural schools Dec. 2021 – Dec. 2022